

Scheduling Algorithm Comparator

Priority vs SRTF . One clear comparison. See who schedules better.

Scenario D) Validation Case :

Process Input 2 processes

PROCESS ID	ARRIVAL TIME	BURST TIME	PRIORITY	ACTION
P1	0	5	2	X
P1	-1	0	-2	X

- Row 2: Duplicate ID "P1".
- Row 2: Arrival Time must be non-negative integer.
- Row 2: Burst Time must be positive integer.
- Row 2: Priority must be non-negative integer.

+ Add Process

Run Simulation

Clear All

1st Fix :

PROCESS ID	ARRIVAL TIME	BURST TIME	PRIORITY	ACTION
P1	0	5	2	X
P2	1	0	-2	X

- Row 2: Burst Time must be positive integer.
- Row 2: Priority must be non-negative integer.

2nd Fix :

Process Input 2 processes

PROCESS ID	ARRIVAL TIME	BURST TIME	PRIORITY	ACTION
P1	0	5	2	X
P2	1	3	-1	X

- Row 2: Priority must be non-negative integer.

+ Add Process

Run Simulation

Clear All

3rd Fix :

Process Input 2 processes

PROCESS ID	ARRIVAL TIME	BURST TIME	PRIORITY	ACTION
P1	0	5	2	X
P2	1	3	3	X

+ Add Process

Run Simulation

Clear All

Scenario (A) Basic Mixed Workload

Process Input 4 processes

PROCESS ID	ARRIVAL TIME	BURST TIME	PRIORITY	ACTION
P1	0	8	2	X
P2	1	4	1	X
P3	2	9	3	X
P4	3	5	2	X

+ Add Process

Run Simulation

Clear All

Priority — Gantt Chart

Preemptive Priority (lower number = higher priority)



Priority — Results

PID	AT	BT	PRI	CT	WT	TAT	RT
P1	0	8	2	12	4	12	0
P2	1	4	1	5	0	4	0
P3	2	9	3	26	15	24	15
P4	3	5	2	17	9	14	9
AVG	--	--	--	--	7.00	13.50	6.00

7.00

AVG WAITING TIME

13.50

AVG TURNAROUND TIME

6.00

AVG RESPONSE TIME

SRTF — Gantt Chart

Shortest Remaining Time First (preemptive)



SRTF — Results

PID	AT	BT	PRI	CT	WT	TAT	RT
P1	0	8	2	17	9	17	0
P2	1	4	1	5	0	4	0
P3	2	9	3	26	15	24	15
P4	3	5	2	10	2	7	2
AVG	--	--	--	--	6.50	13.00	4.25

6.50

AVG WAITING TIME

13.00

AVG TURNAROUND TIME

4.25

AVG RESPONSE TIME

Scenario (B) Conflict Between Priority & Brust Time

Process Input 4 processes

PROCESS ID	ARRIVAL TIME	BURST TIME	PRIORITY	ACTION
P1	0	10	1	X
P2	1	3	3	X
P3	2	2	4	X
P4	3	1	2	X

+ Add Process

Run Simulation

Clear All

Priority — Gantt Chart

Preemptive Priority (lower number = higher priority)



Priority — Results

PID	AT	BT	PRI	CT	WT	TAT	RT
P1	0	10	1	10	0	10	0
P2	1	3	3	14	10	13	10
P3	2	2	4	16	12	14	12
P4	3	1	2	11	7	8	7
AVG	--	--	--	--	7.25	11.25	7.25

7.25
AVG WAITING TIME

11.25
AVG TURNAROUND TIME

7.25
AVG RESPONSE TIME

SRTF — Gantt Chart

Shortest Remaining Time First (preemptive)



SRTF — Results

PID	AT	BT	PRI	CT	WT	TAT	RT
P1	0	10	1	16	6	16	0
P2	1	3	3	4	0	3	0
P3	2	2	4	7	3	5	3
P4	3	1	2	5	1	2	1
AVG	--	--	--	--	2.50	6.50	1.00

2.50
AVG WAITING TIME

6.50
AVG TURNAROUND TIME

1.00
AVG RESPONSE TIME

Scenario (C) Starvation-Sensitive Case

Process Input 5 processes

PROCESS ID	ARRIVAL TIME	BURST TIME	PRIORITY	ACTION
P1	0	20	5	X
P2	1	2	1	X
P3	2	2	1	X
P4	3	2	2	X
P5	4	2	2	X

+ Add Process

Run Simulation

Clear All

Priority — Gantt Chart

Preemptive Priority (lower number = higher priority)



Priority — Results

PID	AT	BT	PRI	CT	WT	TAT	RT
P1	0	20	5	28	8	28	0
P2	1	2	1	3	0	2	0
P3	2	2	1	5	1	3	1
P4	3	2	2	7	2	4	2
P5	4	2	2	9	3	5	3
AVG	--	--	--	--	2.80	8.40	1.20

2.80
AVG WAITING TIME

8.40
AVG TURNAROUND TIME

1.20
AVG RESPONSE TIME

SRTF — Gantt Chart

Shortest Remaining Time First (preemptive)



SRTF — Results

PID	AT	BT	PRI	CT	WT	TAT	RT
P1	0	20	5	28	8	28	0
P2	1	2	1	3	0	2	0
P3	2	2	1	5	1	3	1
P4	3	2	2	7	2	4	2
P5	4	2	2	9	3	5	3
AVG	--	--	--	--	2.80	8.40	1.20

2.80
AVG WAITING TIME

8.40
AVG TURNAROUND TIME

1.20
AVG RESPONSE TIME

Scenario (A) Basic Mixed Workload

Algorithm Comparison

- SRTF has lower average Waiting Time (6.50 vs 7.00).
- SRTF has lower average Turnaround Time (13.00 vs 13.50).
- SRTF has better average Response Time (4.25 vs 6.00).
- SRTF naturally favors short jobs; Priority Scheduling prioritizes high-priority processes regardless of length.

Final Conclusion

Analysis for 4 processes:

SRTF outperforms Priority Scheduling in both waiting and turnaround time. This is expected due to SRTFs optimality in minimizing average completion time.

Starvation note: P3 may experience starvation under Priority Scheduling if higher-priority processes keep arriving. SRTF ensures eventual completion of all processes.

Conclusion: Priority Scheduling is suitable for policy-driven systems where process importance matters. SRTF is ideal for throughput-oriented environments.

Scenario (B) Conflict Between Priority & Brust Time

Algorithm Comparison

- SRTF has lower average Waiting Time (2.50 vs 7.25).
- SRTF has lower average Turnaround Time (6.50 vs 11.25).
- SRTF has better average Response Time (1.00 vs 7.25).
- SRTF naturally favors short jobs; Priority Scheduling prioritizes high-priority processes regardless of length.

Final Conclusion

Analysis for 4 processes:

SRTF outperforms Priority Scheduling in both waiting and turnaround time. This is expected due to SRTFs optimality in minimizing average completion time.

Conclusion: Priority Scheduling is suitable for policy-driven systems where process importance matters. SRTF is ideal for throughput-oriented environments.

Scenario (C) Starvation-Sensitive Case

Algorithm Comparison

- Same average Waiting Time.
- Same average Turnaround Time.
- Same average Response Time.
- SRTF naturally favors short jobs; Priority Scheduling prioritizes high-priority processes regardless of length.

Final Conclusion

Analysis for 5 processes:

Mixed results — the relative performance depends on the correlation between priority values and burst durations.

Starvation note: P1 may experience starvation under Priority Scheduling if higher-priority processes keep arriving. SRTF ensures eventual completion of all processes.

Conclusion: Priority Scheduling is suitable for policy-driven systems where process importance matters. SRTF is ideal for throughput-oriented environments.